

Development and Analysis of a Novel Obstacle Avoidance Strategy for Multi-Robot System Inspired from Bug-1 Algorithm

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Abstract

This paper addresses the development and implementation of an obstacle avoidance strategy for multi-robot system operating in an unknown environment. The novel strategy has been based on conventional Bug-1 obstacle avoidance algorithm which is a non-heuristic method for obstacle avoidance in unknown environment. In Bug-1 algorithm a robot circumnavigate the obstacle to find the co-ordinates of the point having minimum distance to goal. In case of the new strategy, two robots will circumnavigate the obstacle in such a manner that it will reduce both the total travel time and the distance travelled. Information acquired by the individual robots during the circumnavigation is shared across other robots to accomplish the obstacle avoidance in an efficient manner. A theoretical analysis is carried out to show the improvement in travel time and energy expenditure of the robots in implementing the new strategy. Simulation study is conducted to support the theoretical analysis. The robots using this algorithm are found to have a better travel time and less energy expenditure than conventional Bug-1 algorithm. The obtained simulation results are validated with experiments conducted on two robots and results found to be in good agreement.

Keywords: Bug-1 algorithm, Travel time, Energy expenditure, Obstacle avoidance, Multi-robot system

1. Introduction

Intelligent mobile robots are capable of taking decisions based on their working environment. Obstacle avoidance is one of the primary qualities expected for an intelligent mobile robot.

Ranging from nano-robots to deep space exploration rovers, obstacle avoidance in an unfamiliar and unstructured topography has become a challenging and it is a pre-requisite for the mobile robots. There are many algorithms developed to achieve the obstacle avoidance and the robot motion planning. Some of these approaches are presented [1]. Among the different obstacle avoidance algorithms, Bug algorithm [2] is a class of algorithm that has been extensively used to implement obstacle avoidance in robots. Bug algorithm was introduced as a methodology for non-heuristic motion planning in uncertain environments where information about the environment is not fully known [3]. The algorithm does not require prior information about the location of the obstacle for its working [4], [5]. These algorithms are suitable for obstacle avoidance in unmanned vehicles and robots operating in unstructured environments. The information regarding the obstacle is obtained with the help of sensors attached to the robot. Bug-0[6], Bug-1 [7] and Tangent-Bug [8] are some of the early variants of Bug algorithms. Several other variants of Bug algorithm were developed later with improvements in function. IBA [9], K-Bug [10], DH-Bug [11], Bio-Bug [12] etc. are some of the other variants of Bug algorithm. A study regarding the working of different Bug algorithms and their comparisons is provided [13].

The Bug algorithms provide more reliable and predictable results due to the exhaustive search nature of the algorithm. Although developed for obstacle avoidance, the algorithm has been put into other fields also. The exhaustive searching pattern of the Bug algorithm makes it suitable for developing more realistic path planners which imitates a biological organism/human searching the path [14], [15] rather than heuristics based path planning. The Bug algorithm based strategies have been utilized in video games and military surveillance also.

1.1 Comparison of Bug-1 with other algorithms

Among the primary Bug algorithms, Bug-1 assures more predictable and consistent results due to its exhaustive nature. Other Bug algorithms such as Bug-0 [6] and Bug-2 [7] does not require the robot to navigate completely around an unknown obstacle. Hence, there are cases where they either fail to succeed or take much larger path than the Bug-1. Figure 1 shows cases where Bug-1 perform better. Case-1 is where Bug-0 algorithm fails and Bug-1 will succeed. Case-2 is a case where Bug-2 which is designed for reducing the total travel distance will take a much longer path than Bug-1. Hence, Bug-2 can be considered as failed in the given situation. There are cases like a spiral obstacle where the search is very much exhaustive with Bug-1. However, strategy proposed in this paper will perform much better in such scenarios.

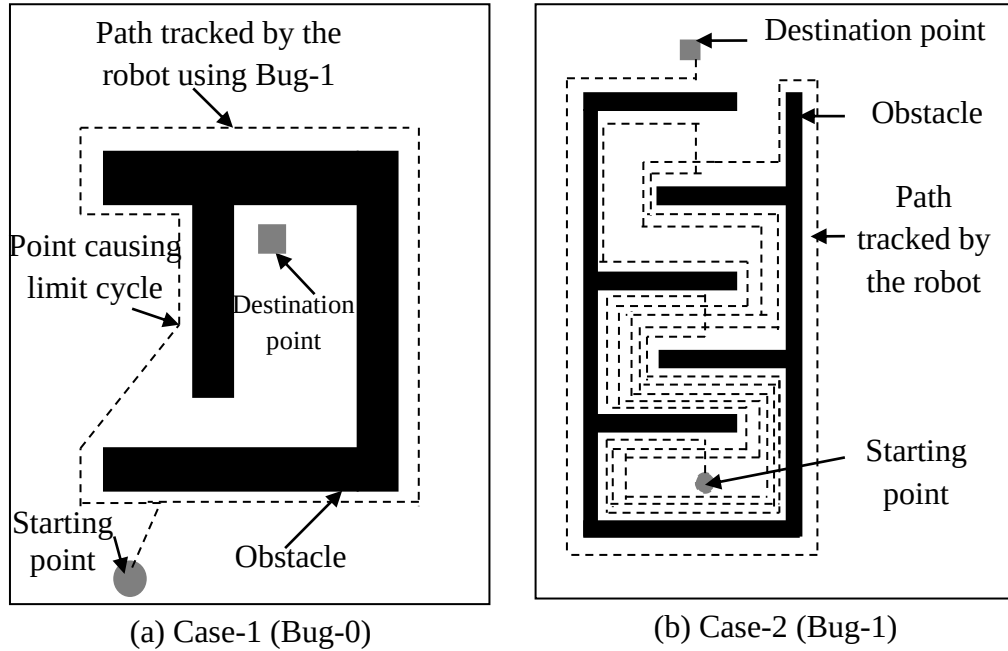


Fig. 1 Scenarios where Bug-1 perform better than other algorithms

There are many other algorithms such as Alg-1, Alg 2, DistBug, IBA, K-Bug, DH-Bug Tangent Bug etc which are derivatives of Bug-1 algorithm. These algorithms although developed as an improvement to Bug-1 may underperform in some cases. Figure 2 shows a case where Bug-1 outperforms Alg-1 and DistBug. A detailed comparison of other

algorithms with Bug-1 is explained [13]. Hence, Bug-1 is selected for the multi-robot strategy presented in this paper. The multi robot strategy presented in the current work will be a reference for developing multi robot obstacle avoidance strategy for other algorithms which are derivatives of Bug-1 algorithm.

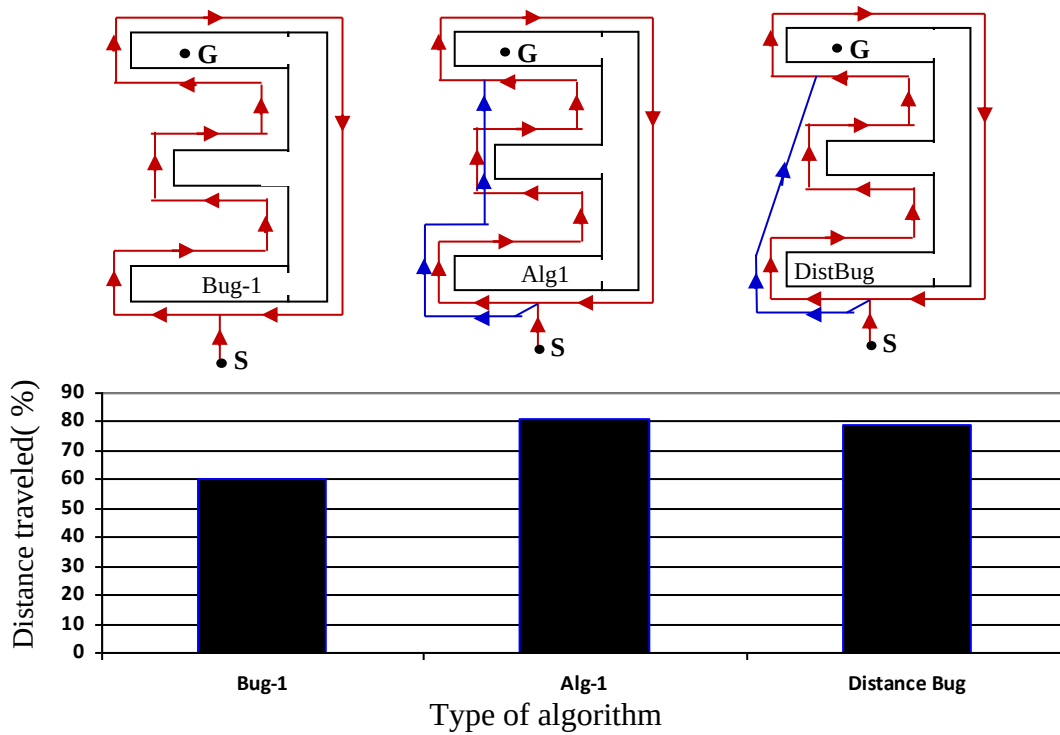


Fig. 2 One of the scenario where Bug-1 outperform other derivative algorithms of Bug-1(Data taken from [13])

Bug algorithms were developed during a period when the focus of robotics was more oriented towards the individual robot that were designed and programmed to execute a specific function. ASIMO [16] and Roomba [17] are examples of such robots. In the recent times, the focus on robotics was shifted from the individual robot to the multi robotic systems [18],[19],[20],[21]. These systems can be used for situations like cleaning, lawn moving and military surveillance [22]. In such systems, large number of robots with limited capabilities coordinates with each other to accomplish a common task. The simple design of the robot make them easy to produce and replace which improves the performance of the multi-robot system [23].

Hence, this paper tries to address the development of an obstacle avoidance strategy for a multi-robot system. The basic vision of the new strategy is to accomplish the obstacle avoidance at a faster rate with more efficient manner in multi-robot systems. Very few research works are available which address the multi robot Bug algorithms. The uniqueness of the current paper is that this is the only paper according to our knowledge, which address multi robot Bug-1 strategy in an unknown map.

The earlier developed Bug-1 algorithm is used as the reference model for the development of this new strategy. Hence, the strategy presented in this paper is a modified Bug-1 algorithm implemented in a multi-robot system such that it will reduce the travel time as well as the energy expenditure of the system. A theoretical analysis is carried out to understand the improvement in travel time and energy expenditure of the system using the proposed strategy. In this analysis the new strategy is compared with Bug-1 algorithm. Both simulation study and experimental study is carried out to verify the results of theoretical analysis. The paper is organized as follows:

Section II details the Bug-1 algorithm and the new strategy developed for multi-robotic system. Section III presents the theoretical analysis of the new strategy and Bug-1 algorithm regarding travel time and energy expenditure. Simulation results for system involving a single robot to six robots are presented in Section IV. Experimental results for a single robot system and two robot systems are presented in Section V.

2. Multi-robot Obstacle Avoidance

Current work attempts to develop Bug-1 based obstacle avoidance strategy for multi-robot system. The Bug-1 algorithm is developed for navigation of a single robot. Figure 3 shows the trajectory of a robot executing Bug-1 algorithm for obstacle avoidance. The robot uses the global knowledge regarding its starting position S and the goal position G to orient itself and move towards the goal. Upon encountering an obstacle at A , the robot starts to

circumnavigate around the obstacle in either clockwise or counter-clockwise direction while checking the distance towards the goal position. The direction of circumnavigation depends on the rule specified by the user. During the circumnavigation, the point having minimum distance to goal position D is recorded and stored. Once the circumnavigation is completed, the robot takes the shortest possible path to the point having minimum distance to goal position G . Upon reaching the point D having minimum distance to goal, the robot continues to move towards the goal location in a straight line path.

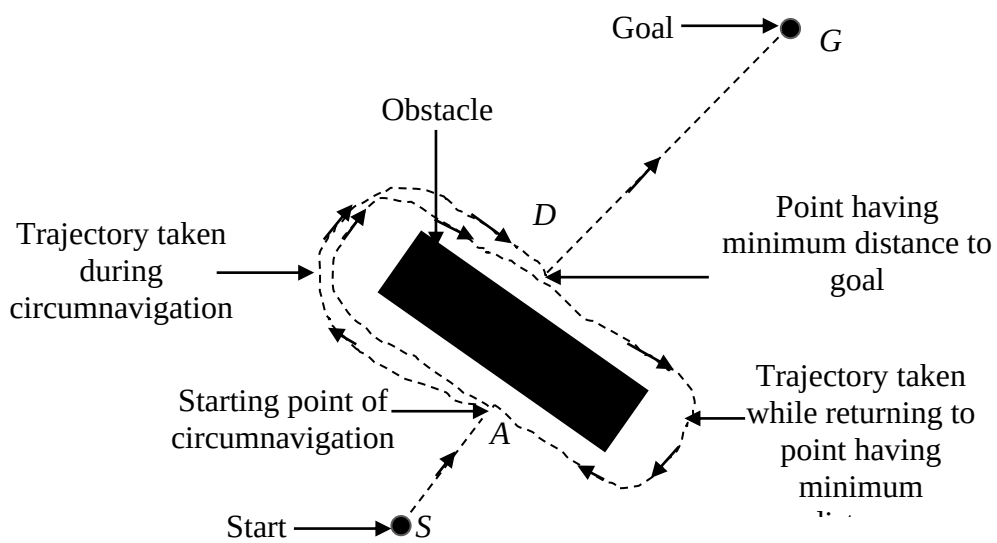


Fig. 3. Trajectory of robot executing Bug-1 algorithm

The multi robot Bug-1 algorithm developed in this work attempts to navigate any number of robots from a given starting position to the final goal position. The algorithm is based on global knowledge acquired by exchanging the information acquired by individual robots by sensing its local environment with the help of the sensors attached to them. In a conventional Bug-1 algorithm when the robot circumnavigates the obstacle, it will find out point having minimum distance to goal. A memory is used to store the co-ordinates of this point and distance from this minimum distance point to goal. In case of Bug-1 based system, this memory serves to find the shortest path once the circumnavigation is completed. In case of a multi-robot system, two or more robots will circumnavigate the obstacle thereby reducing the

total travel time and the distance travelled to circumnavigate the obstacle. Three major decision steps in the new strategy are choice of direction of motion, finding the point having minimum distance to the goal during circumnavigation and choosing the leader and follower robots for further navigation. These steps are explained in the next section.

2.1 Choice of direction of circumnavigation

Initially, the two robots closest to goal are assigned as leader robot and highest priority follower robot respectively (referred further as follower robot). These robots start moving towards the goal from their respective starting positions S_1 and S_2 as shown in Fig. 4.

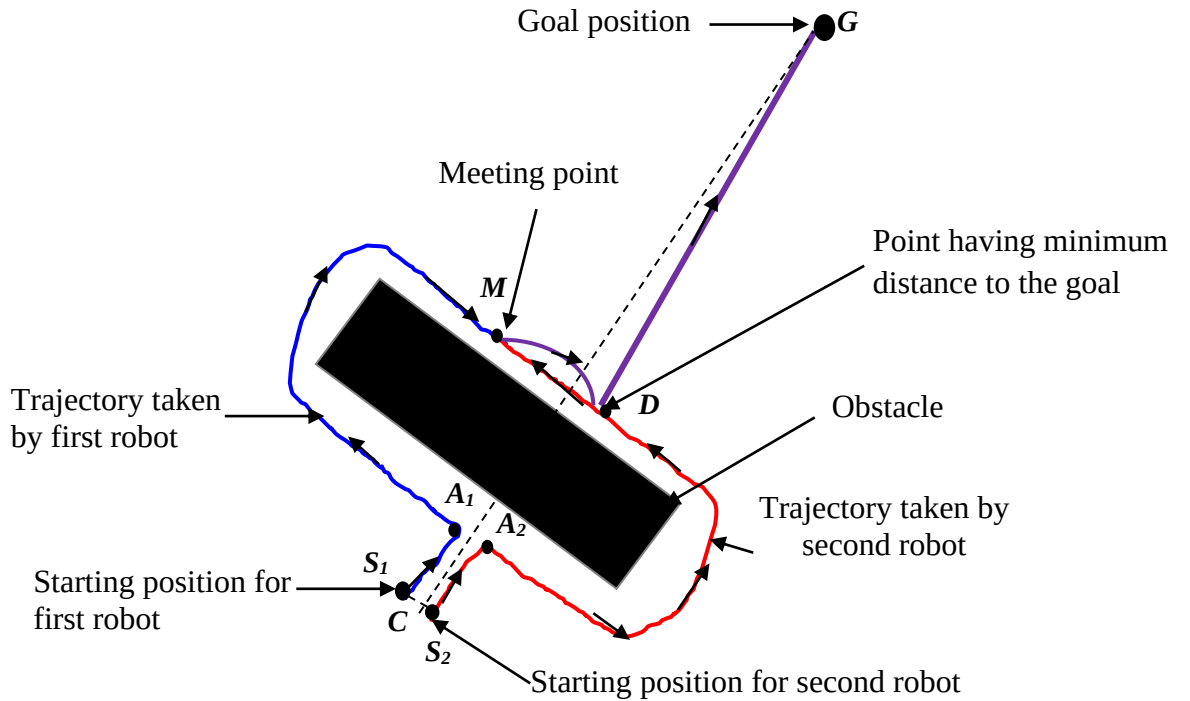


Fig. 4 Trajectory tracked by a two robot system which is using modified Bug-1 strategy

When the obstacle is detected they stop moving. Other robots will also move towards goal in tandem with leader robot and stop their motion when the obstacle is detected. Leader and follower robot encountering the obstacle at A_1 and A_2 respectively has to circumnavigate the obstacle to find the point having minimum distance. The robots can circumnavigate either

clockwise direction or counterclockwise direction. So, the direction of circumnavigation has to be chosen for each of these robots.

In order to choose the direction of circumnavigation, an imaginary line is drawn connecting the start positions S_1 of the leader robot and S_2 of the follower robot. The midpoint of this line is taken as a reference point C . From C another imaginary line is drawn to the goal position G as shown in the Fig. 4. The robot on the left side of this line will act as right sensing robot which circumnavigates around the obstacles in the clockwise direction. On the other hand, the robot on the right side of the imaginary line acts as a left sensing robot and move in the counterclockwise direction around the obstacles. Hence the leader robot and the follower robot move in opposite direction. All other robots will wait till the completion of circumnavigation. Here, it is assumed that all the robots are equipped with sensors in right side as well as in left side. Hence robot can work either as a right sensing robot or as a left sensing robot as situation demands.

2.2 Finding the point having minimum distance to goal

After encountering an obstacle, the leader and the follower start to move around the obstacle in opposite directions that is clockwise and counterclockwise direction respectively following the contour of the obstacle. At regular intervals the location of the robots will be shared with all other robots. As the robots go around the obstacle, they check the distance from their location to the goal location. The coordinates of the point where the distance to the goal is smaller is recorded erasing the previously recorded coordinates. The distance to the goal at this point is also recorded erasing the previously recorded distance. This process continues till the robots meet each other at M .

Since the two robots have taken different routes to reach M , the point having minimum distance to the goal found by these robots will be different. The actual point having minimum distance out of the two points calculated by leader and follower should be determined. For

this, the point having minimum distance and the distance from this point to goal stored by leader robot and follower robot is shared with all other robot. The two points shared by the robots is compared to find the actual point having minimum distance D and the shortest route to this point. This calculation is carried out by all the robots so that every robot will know the point having the minimum distance and the path to the point having minimum distance. After determining the point having minimum distance D , the leader robot and the follower robot start moving towards D . The robot that encountered the point D in its route to M acts as new leader and it turns around and moves towards the point D . The other robot will act as the follower. When the leader and the follower reaches the point having minimum distance D , the rest of the robots starts to move towards D taking the route taken by the robot that recorded this point. Once all the robots reach D , the two robots with the highest energy reserve are assigned the roles of leader and follower respectively.

The flow chart of the algorithm is shown in Fig. 5. One of the primary advantages of this algorithm over conventional Bug-1 algorithm is that it can be applied to any number of robots and as the number of robots increases, the average energy expenditure of the system reduces. Increased number of robots with better energy levels ensures that maximum number of robots reach the goal. Hence, the probability of completing the task increases.

3. Performance Analysis

In order to support the claims of superior performance of the proposed algorithm, a performance analysis of the new strategy has been carried out. The analysis is based on the total travel time and the energy expenditure. In the analysis, the new strategy for multi-robot obstacle avoidance is compared with conventional Bug-1 algorithm. The comparison is carried out in the following manner: theoretical analysis, simulation study and experimental study. Following assumptions are made for the study

1. Only static obstacles are considered for the analysis

2. The robots in the multi robot system is not considered each other as obstacles
3. For simulation, robot is assumed to be a point object
4. Initially, the robots have information regarding their current postures, starting point and destination point only.
5. No information regarding size, shape and position of obstacle in the environment is available to the robot.
6. The robot has minimum sensing capabilities i.e. it can only sense an obstacle near to it using tactile, infrared, ultrasonic sensors or any other similar kind of sensors. So, the robot does not have a capability to make a local map and search for best rout.

Theoretical analysis corresponding to travel time and energy expenditure of the robot is discussed in this section.

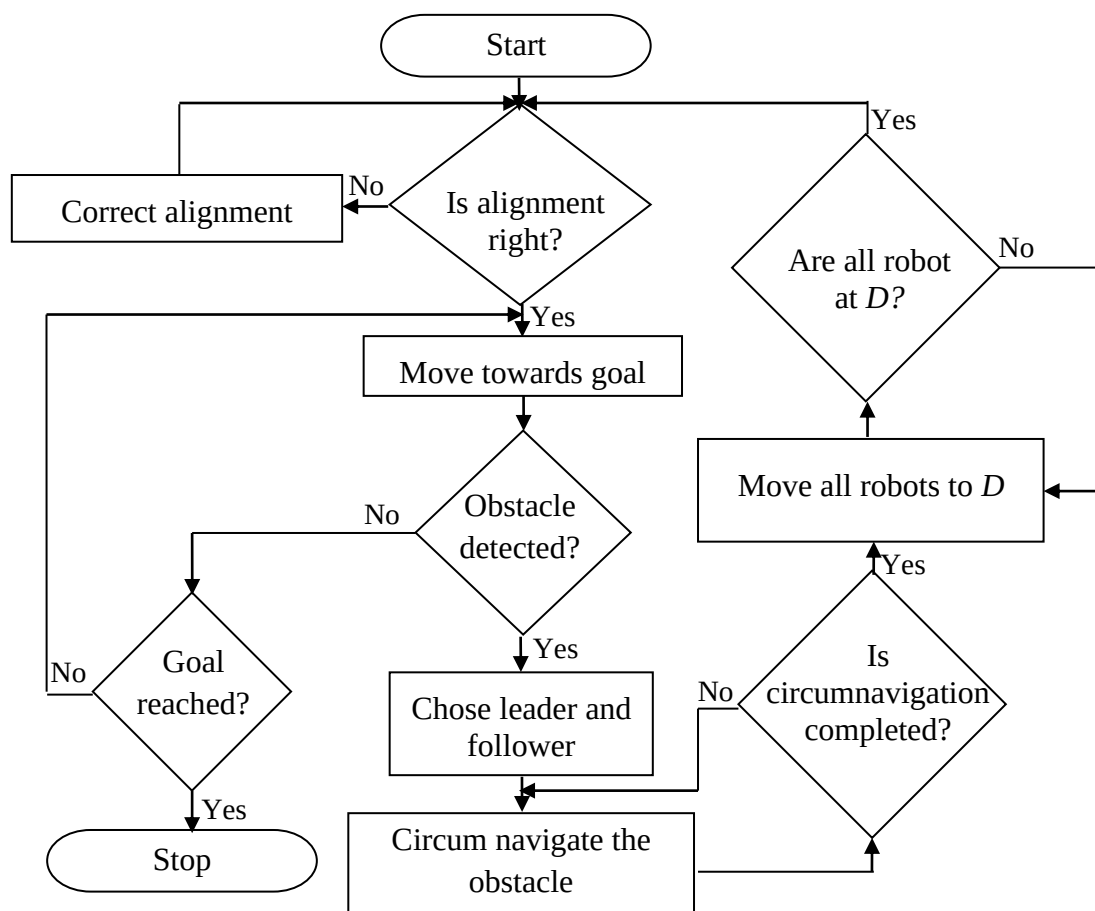


Fig. 5. Flow chart of the new strategy

3.1 Travel Time

A rectangular obstacle is assumed for the theoretical analysis of the algorithm. However, to ensure the generality of the analysis, assume that this rectangular obstacle is enclosed in a circle as shown in Fig. 6. The path taken by the robot is considered to be around the periphery of this circle. In Fig. 6., S , S_1 and S_2 describes the starting positions of the robots while A , A_1 and A_2 are the initial points of contact of the robot with the obstacle, M is the meeting point of the robot while they circumnavigate, D is the point having minimum distance to goal and G is the goal point. In case of single robot system there is no meeting point as there is only one robot. But, the point corresponding to meeting point of multi robot system is marked as M in Fig 6 (a). The area enclosed by the tangents from S in Fig.6 (a) and S_1 and S_2 in Fig.6 (b) to the circle that encloses the obstacle provides the possible range of the locations of the goal if the robot is to encounter an obstacle en-route to the goal point G . For the analysis, goal is assumed to be in this region enclosed by the tangents. Since the robot has to move from meeting point to the point having minimum distance before proceeding to the goal, it can be concluded that the travel time increases as the distance between the meeting point M and the point having minimum distance to goal D increases. It can also be understood from Fig. 6 (a) that for a single robot system, the travel time is maximum if the goal point is located at the midline of the region enclosed by the tangents. For multi robot system from Fig. 6 (b), the travel time will be maximum when the goal location is along the boundary of this region.

For a single robot system, assume T_{SA} as the time taken for the robot to travel from the starting position S to reach the initial point of obstacle detection A . The time taken for the robot to go around the obstacle and return to A is taken as T_{1AA} . Time required for the robot to reach D from A taking the shortest route is T_{1AD} and time for it to travel from D to G is T_{DG} .

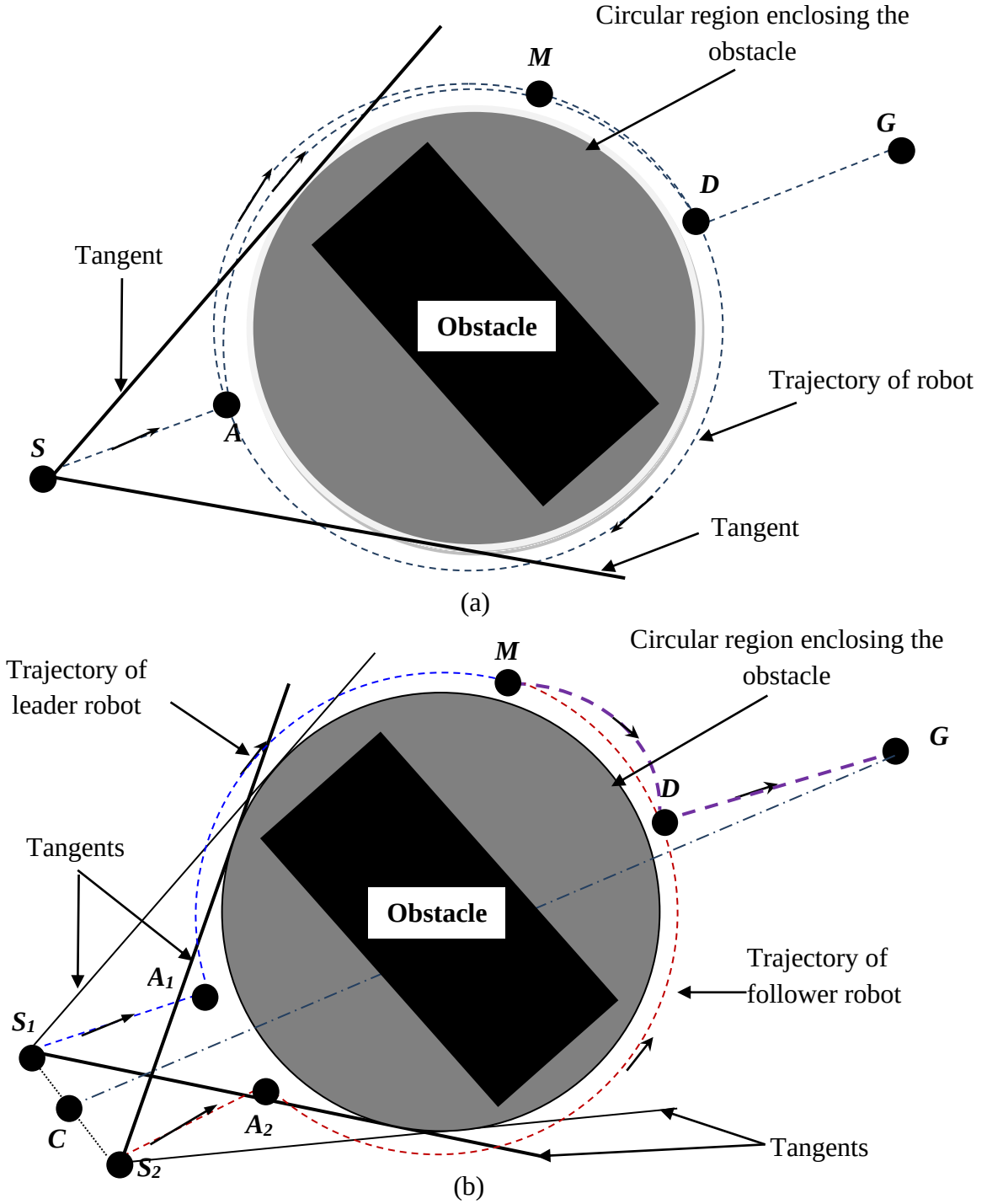


Fig. 6. Comparison of trajectories (a) Bug-1 algorithm (b) Multi-robot obstacle avoidance strategy

In case of multi robot system, T_{SA} is the time taken for the leader and follower robots to travel respectively from their starting position S_1 and S_2 to reach the initial points of contact A_1 and A_2 . It is assumed that the time taken to detect the obstacle is same for single robot as well as

for two robot system. The time duration for the two robots to circumnavigate and meet at M after leaving A_1 and A_2 be T_{2AM} , the time required for the robots to reach D from M be T_{2MD} . It is assumed that T_{1AD} as the time taken for other robots to reach the point having minimum distance. Finally, the time required for all the robots to move from D to the goal location G is taken as T_{DG} .

The total time taken by the two systems are compared by considering $T_{total-I}$, the time taken by the single robot system given by Eq.1 and $T_{total-M}$, the time taken by the multi-robot system given by Eq.2. It is the sum of time taken to reach initial point of contact with obstacle, time taken for circumnavigation, time taken for leader and follower to reach the point D, time taken for all other robot to reach point D from initial point of contact with obstacle and time taken for all robot to reach G from D. So, For the forthcoming calculations, total time refers to the time required by the robot to move from the start position to the target position in the single robot system and the time taken by the final robot to reach the target position in the multi-robot system. To simplify the calculations, the robots are assumed to move with constant velocity.

$$T_{total-I} = T_{SA} + T_{1AA} + T_{1AD} + T_{DG} \quad (1)$$

$$T_{total-M} = T_{SA} + T_{2AM} + T_{2MD} + T_{1AD} + T_{DG} \quad (2)$$

In multi robot system two robots go around the obstacle, the time to meet each other will always be less than or equal to half of the time required for circumnavigation. This is given by Eq.3.

$$T_{2AM} \leq \frac{T_{1AA}}{2} \quad (3)$$

Once the robots meet each other, they will travel back to the point having minimum distance to goal. The point having minimum distance to goal is always located enroute to the meeting point M from the contact point A . Hence, the time taken to travel back to the point having

$$T_{2MD} < T_{2AM} \quad (4)$$

minimum distance to goal will be less than the time required for meeting each other. This is given by Eq.4.

Combining Eq. 3 and Eq. 4

$$T_{2AM} + T_{2MD} < T_{1AA} \quad (5)$$

Eq. 6 can be obtained by comparing Eq. 1 and Eq.2 by taking into account Eq. 5

$$T_{total-I} > T_{total-M} \quad (6)$$

Hence it can be concluded that the time taken for single robot system to reach the goal is larger than the time required for a multi robot system.

3.2 Energy Expenditure

One of the most important aspects of any system is the energy usage. In a system involving multiple robots, an effective form of energy usage ensures better chance of task completion. In the following section an attempt has been made to prove that the average energy expenditure of the multi-robot system as well as total energy expenditure of any robot of multi-robot system implementing the proposed algorithm is always lower than that of a robot executing the Bug-1 algorithm.

In a robot, there are several elements which consume energy such as motors, sensors, data transmission and reception, data processing etc. Since calculation of energy consumed by each element is a tedious process, it is assumed that all these energy divisions are in direct relation to the distance covered. This provides a reasonably a good approximation since the robot spend most of its energy for its motion related operations.

3.2.1. Average energy

Consider $E_{total-I}$ and $E_{total-M}$ as the total energy expenditure for single robot system executing Bug-1 algorithm and two-robot system executing the proposed algorithm respectively. The total energy expenditure $E_{total-I}$ and $E_{total-M}$ are calculated in Eq.7 and Eq. 8 respectively. For

obtaining these equations, the distance travelled by the robot is calculated and then multiplied with a constant k . This constant is used as a conversion factor to calculate the value of energy expenditure from the distance travelled. This is in tandem with the assumption that energy expenditure is in direct relation to the distance covered.

$$E_{total-I} = k \left(SA + \left(\text{Length of curve} (AM + MA + AD + DG) \right) \right) \quad (7)$$

$$E_{total-M} = k \left(SA_1 + SA_2 + \left(\text{Length of curve} (A_1M + A_2M + 2MD + 2DG) \right) \right) \quad (8)$$

To avoid the effect of starting location of the robots on the calculations, it is assumed that $SA = SA_1 = SA_2$. This results in Eq. 9.

$$E_{total-M} = k \left(2SA + \left(\text{Length of curve} (A_1M + A_2M + 2MD + 2DG) \right) \right) \quad (9)$$

Average of energy spends by the two systems E_{avg-I} and E_{avg-M} are calculated in Eq. 10 and Eq. 11 by dividing the total energy expenditure with number of robots in the system.

$$E_{avg-I} = E_{total-I} = k \left(SA + \left(\text{Length of curve} (AM + MA + AD + DG) \right) \right) \quad (10)$$

$$E_{avg-M} = \frac{E_{total-M}}{2} = \frac{k \left(2SA + \left(\text{Length of curve} (A_1M + A_2M + 2MD + 2DG) \right) \right)}{2} \quad (11)$$

$$E_{avg-M} = k \left(SA + \left(\text{Length of curve} \left(\frac{A_1M + A_2M}{2} + MD + DG \right) \right) \right) \quad (12)$$

Comparing Eq. 10 and Eq. 12 it can be understood that average energy of the single robot system will be lesser if and only if Eq. 13 is satisfied. Considering the worst case scenario it is assumed that AD is zero in case of single robot system. Then

$$\frac{A_2M + A_1M}{2} + MD \leq AM + MA \quad (13)$$

From Fig 6 (a) and (b) it can be observed that,

$$AM + MA \geq A_1M + A_2M \quad (14)$$

$$MD < A_1M \quad (15)$$

$$MD < A_2M \quad (16)$$

Equation 17 can be obtained by combining Eq. 14 with Eq.15 and Eq. 16

$$MD < \frac{A_2M + A_1M}{2} \leq \frac{AM + MA}{2} \quad (17)$$

Equation 18 is obtained by taking $(A_2M + A_1M)/2$ to the left side of Eq.17.

$$\frac{A_2M + A_1M}{2} + MD < AM + MA \quad (18)$$

Eq. 18 shows that the Eq. 13 is satisfied, hence

$$E_{avg-M} < E_{avg-I} \quad (19)$$

From the above analysis it is concluded that the average energy expenditure in a two robot system implementing the new strategy is less when compared with the single robot system implementing the Bug-1 algorithm. In a system having more than two robot executing the new strategy, all robots other than the leader robot and follower robot will take shortest path to the point **D**. Hence the energy spend by these robots will be less than that is spend by leader robot or follower robot. This leads to further smaller average energy value if other robots are also considered for calculation of average energy expenditure. Hence, the irrespective of the number of robots involved in the multi robot system, the average of the energies spend by the robots will be lesser in case of multi robot system executing the new strategy.

3.2.2. Total energy expenditure

To compare the total energy of the two systems, the paper shall establish some criteria. The assumptions referred to in the beginning of this sections extends to this section. Further, since it's a comparison of total energy of two systems, the paper shall assume equal number of

robots in both systems and that all the robots and their starting positions are completely similar.

It can be seen that in either system mentioned above once the minimum distance to target point is identified, the robots that follow shall take the same path – directly move to the point of minimum distance once obstacle is encountered and then to target position. For the conventional Bug-1 algorithm based system, this happens from the first follower robot or the second robot to encounter the obstacle. While the same happens with the second follower or the third robot in the system to encounter the obstacle. Hence, it can be seen that to compare the total energy of two systems, one executing conventional Bug-1 and the other executing the modified Bug-1 strategy proposed by the paper, it is sufficient to compare the total energy of the two systems with each consisting of two robots as the rest of the robots would have similar energy expenditure. This energy expenditure of the robots that follow after the minimum distance to the point is identified is given by Eq. 20.

$$E_{fol} = k(SA + \text{length of curve}(AD+DG)) \quad (20)$$

For a two robot system executing the conventional Bug-1 algorithm, the total energy denoted by $E_{total-C}$ will include the additional energy from one follower robot and is identified by Eq. 21.

$$E_{total-C} = E_{total-I} + E_{fol}$$

$$E_{total-C} = 2k(SA + \text{length}(AD+DG)) + k(\text{length of } MA+AM) \quad \text{--- (21)}$$

It can be observed that total energy of the conventional BUG algorithm robot system will be lesser if and only if Eq. 22 is satisfied.

$$E_{total-C} \leq E_{total-M} \quad \text{---- (22)}$$

For simplification, we shall assume $SA=SA_1=SA_2$ as before to yield Eq. 23

$$2k(SA + \text{length}(AD+DG)) + k(\text{length of } MA+AM) \leq 2k(SA + \text{length}(2MD+2DG+MA_1+AM_2)) \quad \text{---(23)}$$

Removing common terms from both sides, we have

$$\text{Length of}(AD) + \text{length of}(MA+AM)/2 \leq (\text{length of}(A_1M + A_2M))/2 + MD$$

With minute approximation, we take $\text{length of}(MA+AM) \approx \text{length of}(A_1M + A_2M)$ to give us the final criterion for $E_{total-C}$ to be less than $E_{total-M}$ for a two robot system, given by Eq. 24.

$$\text{Length of } (AD) \leq \text{Length of } (MD) \quad \text{---(24)}$$

From figure 4, it can be observed that M must always lie in between A and D, hence a case where length of path AD being less than the length of path MD is not possible. So we conclude that for two systems consisting of equal number of robots, the modified BUG algorithm proposed by this paper shall consume less energy in total.

3.2.3. Energy expenditure of any robot

Consider a multi-robot system, if there is more than one obstacle and more than two robots, the initial leader-follower set of robots turn idle at the end of circumnavigation of the first obstacle and another set of robots with higher energy reserve take up the roles of leader and follower when second obstacle is detected. It is possible to conclude from the above discussion that the energy reserve of individual robots in a system with more than two robots will be higher in comparison to a two-robot system where there is no other robot available to take the role of leader and follower. So, the worst case scenario of energy expenditure occurs to the system consisting of just two robots. Hence, in the following section the energy of individual robots in a two robot system is compared to that of conventional Bug-1 system to show that the multi-robot system will have lower energy expenditure for individual robots also.

Similar to calculation of average energy expenditure, the distance covered is considered to be the parameter which determines the energy expenditure of individual robots. The total energy expenditure of a conventional robot system $E_{total-I}$ provided by Eq.7 is compared with $E_{total-M1}$ and $E_{total-M2}$, the individual energy expenditure of leader and follower robots in two robot system calculated by Eq.20 and Eq.21 respectively.

$$E_{total-M1} = k(SA_1 + \text{Length of curve}(A_1M + MD + DG)) \quad (20)$$

$$E_{total-M2} = k(SA_2 + \text{Length of curve}(A_2M + MD + DG)) \quad (21)$$

Equation 22 is arrived with respect to Fig 6 (a) and (b), assuming $A_1M > A_2M$

$$AM + MA > A_1M + MD > A_2M + MD \quad (22)$$

Considering Eq. 22 in EQ.7, Eq.20, Eq.21 and assuming length of curve $AD = 0$ and $SA_1 = SA_2 = SA$,

$$E_{total-I} > E_{total-M1} > E_{total-M2} \quad (23)$$

The above equations verify that the energy expenditure of a single robot executing Bug-1 algorithm is always greater than that of individual robots in the dual-robot system executing the new strategy. The significant observation is that the final energy reserves of any robot in a multi-robot system will always be greater than that of a Bug-1 algorithm based robot executing the same task.

If the number of robots involved in the two systems are different, by considering net energy for the analysis, one cannot conclusively say whether net energy will be higher or lower for the new algorithm. In some cases, net energy will be less in other cases it may be more. It depends on distance from start to goal as well as size and shape of obstacle. In a two robot system, the net energy will be lower for the new method if the goal is very near to meeting point and distance to goal is comparatively less than the circumference of the obstacle. In case of a comparison between a single robot navigating the Bug-1 algorithm and a multi robot system with more than two robots, the net energy used will be more for multi robot system executing the new algorithm. The above statement holds good when number of robots in the system is one in case of Bug-1 and more than one in case of new strategy. In a multi-robot obstacle avoidance scenario, when the system executing Bug-1 and the system executing the new algorithm has same number of robots, the net energy used will always be less for the system using the new algorithm. However, in all cases irrespective of the number of robots involved, the final energy reserve of each of the robot in the system executing the new strategy will be higher than the final energy reserve of a system executing the Bug-1.

This in turn leads to the conclusion that the proposed algorithm is more energy efficient than the conventional algorithms.

Similarly, if the net distance travelled is considered for analysis, one cannot conclusively say whether net distance travelled by the robot using the new algorithm will be higher or lower when compared with the Bug-1 algorithm.

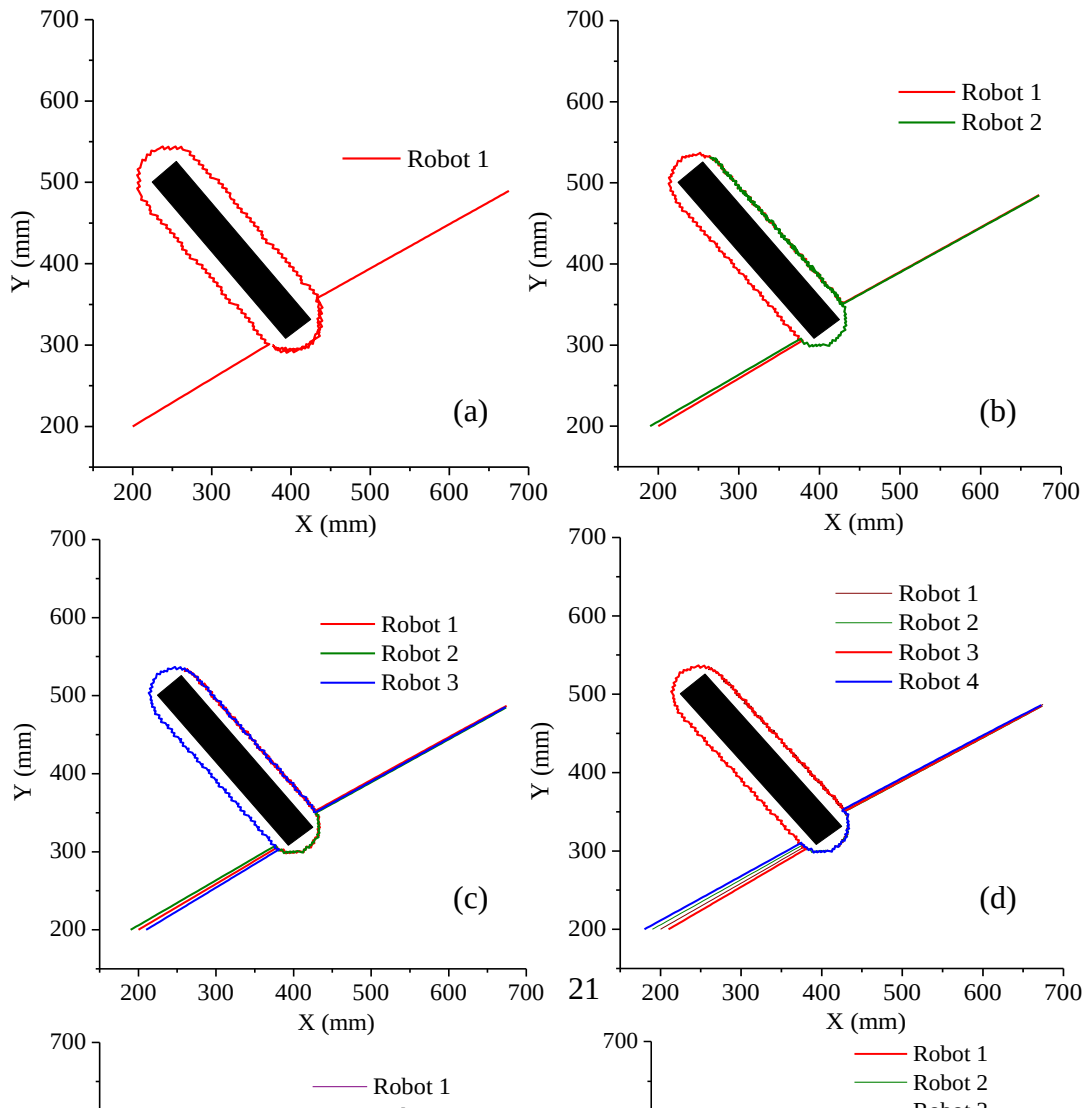
4. Simulation

The theoretical results obtained in the previous sections favors the proposed strategy over existing Bug-1 algorithm. A simulation study has been carried out to verify the same. For the simulations, MATLAB v 8.6 was utilized. In the simulations, the distance covered and the time taken for navigation across given obstacles is studied for systems involving different number of robots. The numbers of robots in the system considered in simulation consist of one robot up to six robots. So, total of six simulations are carried out. Each of the simulation is carried out three times to confirm that the result are not varying more (± 5 mm in case of distance, ± 0.25 s in case of time of travel). For the simulation, the robot was assumed to be a point sized object with object sensing capability till a finite distance in eight equally distributed directions at 45° , first one being along the direction that the robot faces. Initial position was set near to (200,200) with the robot facing the direction of X-axis ie. 0 degrees.

Fig. 7 (a) to Fig. 7 (f) shows the simulation results for path tracked by one robot to six robots. The path tracked by single robot system shown in Fig .7 (a) is based on conventional Bug-1 algorithm. This is used as a reference for comparison of performance of the other multi-robot simulations which executes the proposed multi-robot obstacle avoidance strategy. During the simulation the distance travelled by each robot and the travel time involved are recorded. A common goal coordinates (700,500) is set for all the simulation. A rectangular obstacle is placed along the line joining the points (250,475) and (400,300). The obstacle is inflated to compensate the size of the robot which is 70 mm in diameter. The starting position of the

robots are set at (200,200) for 1st robot, (190,200) for 2nd robot, (210,200) for 3rd robot, (180,200) for 4th robot, (220,200) for 5th robot and (190,190) for 6th robot.

The simulation results are tabulated in Table 1. The total travel time is plotted in Fig. 8 (a) against the number of robots. Similarly, in Fig. 8 (b) average distance covered by the robots is plotted against the number of robots involved. It is observed from the obtained results the travel time is less for two robot system than for all other system. For system involving more than two robots, a slightly increasing trend is observed in travel time. This is majorly due to the extra travel time required for all other robots to reach the point having minimum distance to goal. Starting position of the robots which are selected as the leader and follower robot for circumnavigation also plays a key role in determining the travel time. In five robots as well as in six robot system 5th robot and 3rd robot are selected as the leader and follower robots. This resulted in almost same travel time in these systems. However in all the cases, travel time was less than that of single robot system.



It was observed that the travel time of the robots using the new strategy is at least 11 % less when compared with the robot using conventional Bug-1 algorithm. The difference was not very large since the meeting point of the robot is far away from the point having minimum distance to goal which is the worst case scenario in case of the modified Bug-1 algorithm. The result would have been much better if these points are closer. Hence for experimentation the obstacle is arranged in such a way that the meeting point is closer to the point having minimum distance to goal. It

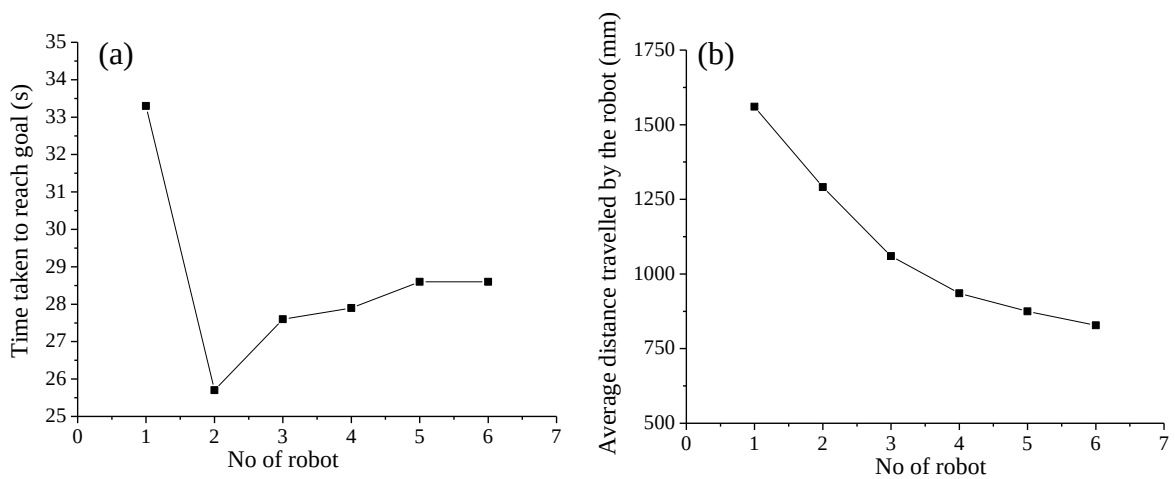


Fig. 8. Graph showing trend of (a) Total time taken to reach the goal by each robot
(b) Average distance travelled by the robot

Table 1. Simulation result

Number of robots	Time taken to reach the goal (s)	Average distance travelled by the robot (mm)	Maximum distance travelled (mm)
1	33.3	1560.0	1560.0
2	25.7	1290.8	1295.8
3	27.6	1060.0	1325.0
4	27.9	935.5	1306.4
5	28.6	874.5	1306.4
6	28.6	827.7	1306.4

The average distance travelled by the robot is found to be decreasing with increasing number of robots. Thus, it is expected that the average energy expenditure will also decrease as number of robot increases. It can also be observed from Table 1 that the robot in single robot system travelled the highest distance. This result support that the energy expenditure of any robot in a multi-robot system with the new control strategy will be less than that of robot implementing the Bug-1 algorithm. Thus, the simulation result supports the theoretical analysis carried out in the previous section. Similarly, the maximum distance travelled by any of the robot in the multi robot system using the new strategy is found to be less than that of the robot using Bug 1. So, after avoiding the obstacle, all of the robots in the multi robot system will have more energy reserve than single robot system. If the net distance traveled by the robot is considered, a multi-robot system using the new strategy will have more net distance travelled. However, it has to be noted here that, each of the robot in the multi robot system will have its own energy reserve (battery). Hence, it is fair to compare maximum distance travelled by the robot rather than net distance travelled by the system.

When it comes to a scenario where multi robot system with more than two robots avoiding multiple obstacles, there is an added advantage with the algorithm developed in the current work. Figure 9 shows a scenario where multi robot system is avoiding multiple obstacles. If

there are only two robots in the system, then these two robots are only available for circumnavigating both the obstacles. This scenario is shown in Fig. 9 (a). Whereas, if there are more than two robots, say 4 robots and two obstacles in the path, the first two robot will be used for circumnavigating the first obstacle. Third and fourth robot will be utilized for circumnavigating the second obstacle as shown in Fig 9 (b). Thus, in this case there are lesser chances for any robot in the system to run out of its energy reserve (battery). Hence, there is a higher chance for all the robots in the system to reach the goal without running out of fuel. This is not possible in case of a multi robot system which is using Bug-1 strategy.

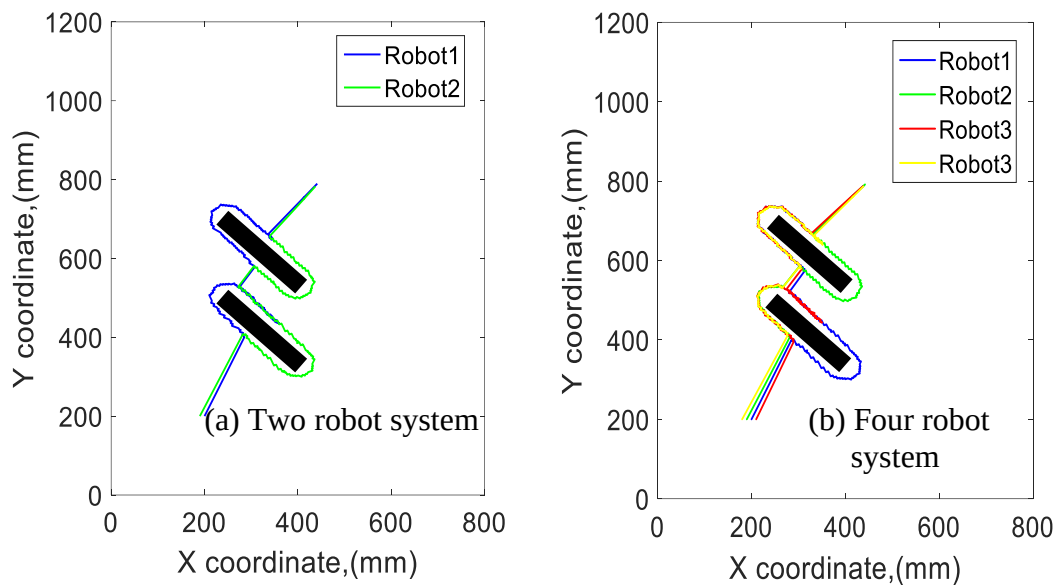


Fig. 9. Multi robot system avoiding multiple obstacles in the path using the new strategy
(a) Two robot system (b) Four robot system

Position of Goal	Number of robots	Time taken to reach the goal (s)	Average distance travelled by the robot (mm)	Maximum distance travelled (mm)
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Outside obstacle	1	136	5907.1	5907.1
	2	93	4457.3	4499.7
	3	115	3438.8	4499.7
Inside obstacle	1	165	7042.5	7042.5
	2	61	2946.8	2962.7
	3	119	2920.3	2962.7

Figure 10 shows the simulation result of robot avoiding a complex obstacle using the Bug-1 strategy as well as new strategy. The simulation results are tabulated in Table 2. This result also support the previous claim of having lower travel time and lower travel distance for the new algorithm.

Table 1. Results of robot avoiding the complex obstacle

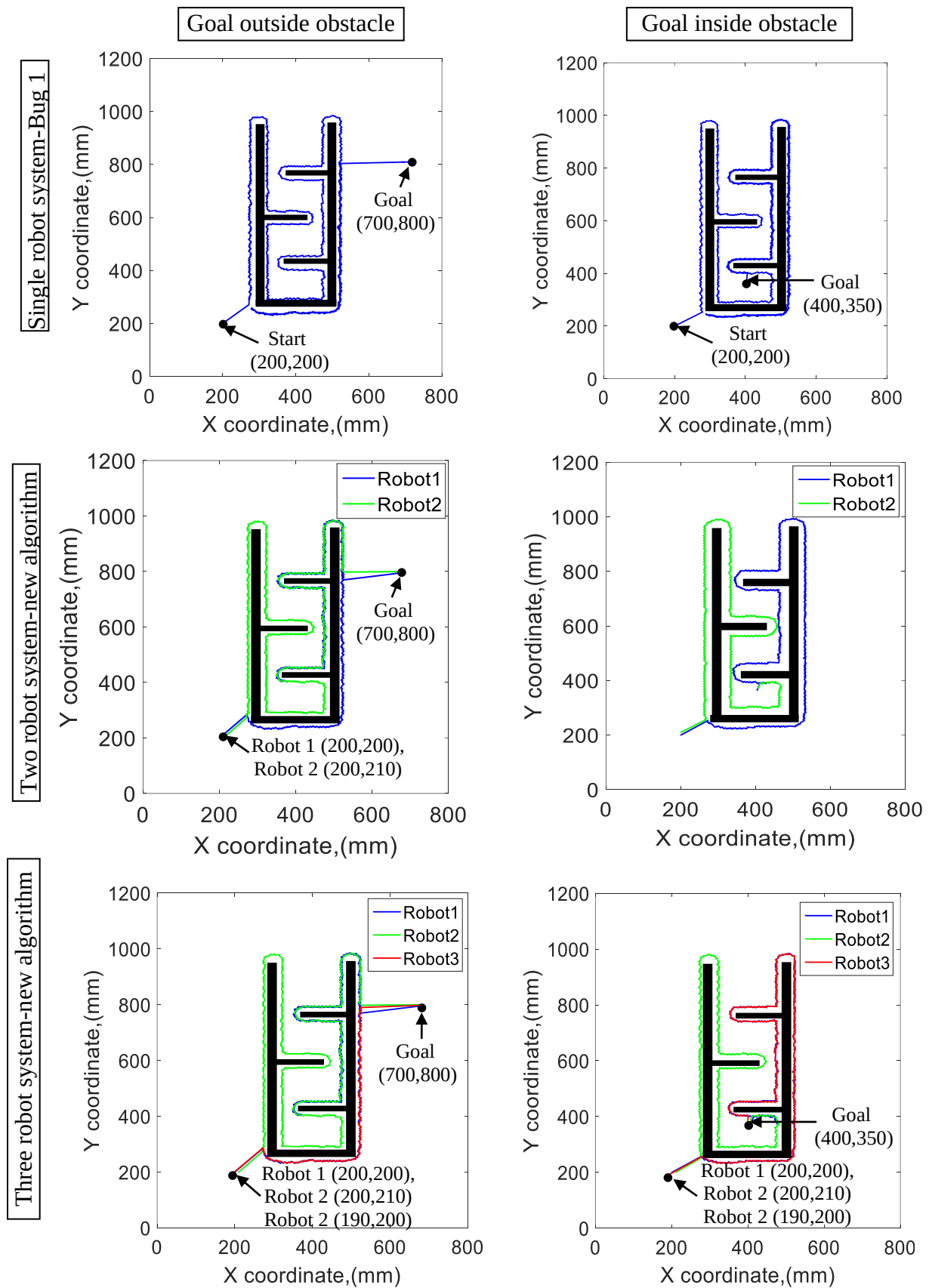


Fig. 10. Simulation result of robot avoiding a complex obstacle

5. Experimental Results

The experimental setup consists of robots running on a platform of dimensions 1600 mm x 800 mm as shown in Fig. 11. A vision based measurement system which is developed in house is utilized for measuring the position of the robot during experiment. The position information is further used to find the trajectory tracked by the robot. The measurement obtained through this measurement system is presented as experimental result. However, the robot is not using information from this measurement system as a feedback instead it is relying on the odometry to find its posture. So, vision based navigation or any information obtained through the vision system is not utilized in the experiments for controlling the robot. For enabling this vision based measurement, the platform was illuminated in such a way that light falls evenly on the two-dimensional platform. A webcam placed on the ceiling is utilized to capture video of the experiment. The video captured by the webcam is send to an image processing system. The path tracked by the e-puck robot is identified using image processing. It is also used to shed light on the velocity of the robots while moving from starting position to the goal position.

In order to verify the findings from simulation results, the algorithm is implemented into e-puck robot which is a wheeled mobile robot used for educational purposes. The robot is made to navigate from a starting position to a goal position on the experimental platform. The starting position is at (200,200) for 1st robot and (120, 200) for 2nd robot and the goal is at (700, 500). A rectangular obstacle is placed along the line joining the points (250,500) and (400,75). Obstacle is placed midway between the starting position and the goal position to ensure that it is detected by the robot during the experiments.

For experiments e-puck robot is used, which is programmed in embedded C using MPLAB V 8.92 IDE from microchip technologies. The programming is done in such a way that the

robot can be controlled from a MATLAB environment. Hence, for both simulation as well as for experimentation MATLAB was used.

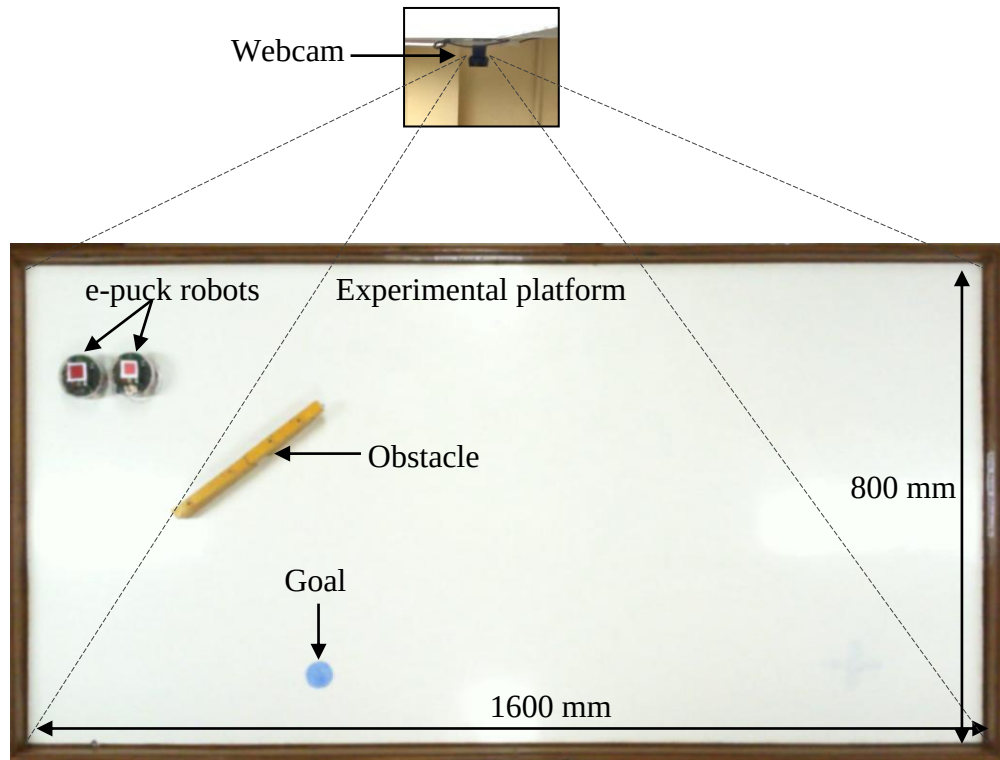


Fig. 11. Experimental setup

The experiment is divided into two parts. Experiment 1- The conventional Bug-1 algorithm is programmed into a single e-puck robot and the robot is made to move from the start position to goal position. Experiment 2 - The algorithm proposed in the current work is implemented on two e-puck robots and the robots are made to move from the start to goal position. Each of these experiments are carried out two times. The general description and results of the experiment are explained in the following section.

5.1. Experiment 1 : Bug-1 algorithm

In this experiment, an e-puck robot is programmed with the Bug-1 algorithm and is made to navigate to the goal location from the start position. It is assumed that the robot reached the goal if robot is within 30 mm of the goal location. This tolerance is provided since exact

attainments of goal location require extensive control algorithms and calibration. The trajectory tracked by the e-puck robot as well as velocity of the robot obtained with the help of image processing system is shown in Fig. 12. The trajectory of the robot can be divided into four parts as shown in the Fig. 12 (a) and are discussed below. The velocity profile is also divided accordingly as shown in Fig. 12. (b).

Section (I): The robot is idle till the tenth second waiting for start command from the user. Then the robot align itself facing the goal and starts moving towards the goal with constant linear velocity. After aligning itself with goal direction, the robot starts moving towards the goal with a velocity of 53 mm/s till it encounters an obstacle. The detection of obstacle results in the sudden velocity dip near thirteenth second.

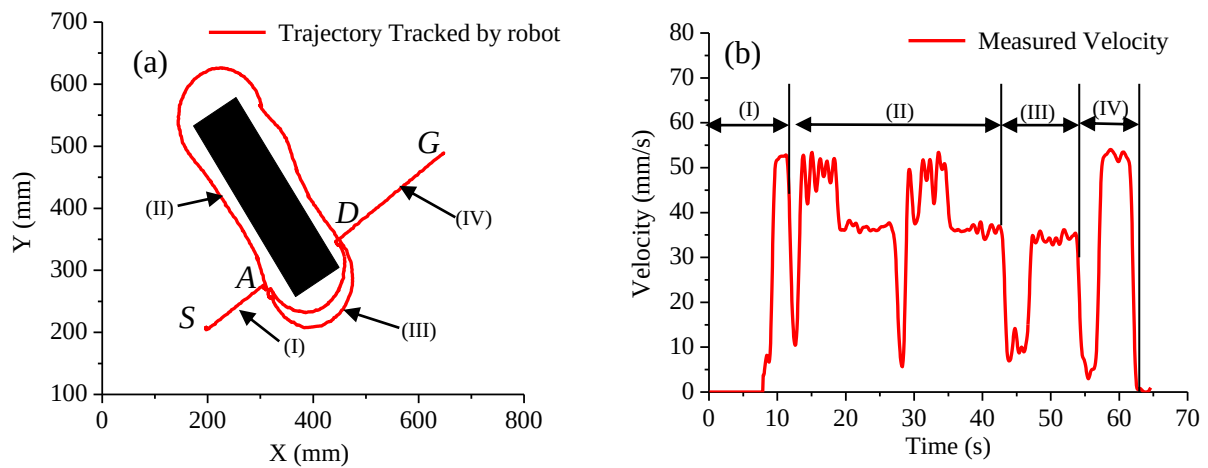


Fig. 12. Bug-1 algorithm (a) Trajectory of the robot (b) Velocity of the robot

Section (II): The robot detects an obstacle and starts circumnavigating it in the clockwise direction. For this, robot orients itself parallel to the obstacle contour. This causes an initial dip in the velocity profile. Then it starts moving along the perimeter of the obstacle at a velocity of 53 mm/s till the corner of the obstacle. It again lowers its velocity while going around the corner to 35 mm/s as displayed by the velocity profile. This lower velocity allows the robot to turn smoothly. The sudden dips in the velocity along the Section (I) as well as in

Section (II) are due to course correction to maintain a reasonable distance from the contour of the obstacle as the robot moves around the obstacle.

During circumnavigation, robot records the point having minimum distance to the goal and the distance to goal this point. The total distance travelled from the point of obstacle detection is also recorded. Once the robot reaches within 30 mm of the point of initial detection of obstacle, the circumnavigation is stopped. The robot reaches to this point at 43rd second. Then the robot checks for the shortest route to the point of minimum distance to the goal.

Section (III): Once the circumnavigation is complete, the robot chooses the shortest path to reach the point of minimum distance to goal. In the present scenario, the robot chooses the path in anticlockwise direction as the shortest path. The robot moves through this path to reach the point having minimum distance to goal. For this, the robot reorients itself for taking anticlockwise direction of traversal. The initial lower velocity in this section corresponds to this reorientation. Then it moves towards the point having minimum distance to goal with a velocity of 35 mm/s following the contour of the obstacle. The velocity is lower in this case as the robot is going around the corner. The robot reaches the point having minimum distance to goal at fifty fifth second.

Section (IV): When robot reaches the point having a minimum distance to goal, it orients itself in the direction of the goal. This corresponds to lower velocity in the beginning of this section. Then the robot starts to move in the direction of the goal at a velocity of 53 mm/s and constantly checks the distance to the goal as well as any obstacle in the path. It is considered that the goal is attained if the goal is within 30 mm. From the velocity profile it is observed that at 63 s the robot reaches the goal. The total travel time of the robot to reach the goal, neglecting the initial waiting time is 54 s and the maximum velocity attained is 53 mm/s. The total distance travelled by the robot while executing the Bug-1 is 2440 mm.

5.2. Experiment 2 : Multi-robot obstacle avoidance strategy

Two e-puck robots are programmed with the modified Bug-1 algorithm and are made to navigate to the goal location (700, 500) from the starting position (200, 200) and (120, 200). The image processing is utilized to shed light on the trajectory tracked by the robots and velocity of the robot. Since there are two robots involved, the plots are developed for both the robots separately. Fig. 13 shows the path tracked by the robots during the experiment. Similar to the Bug-1 experiment, the trajectory and velocity profile of the robot is divided into four sections as discussed below:

Section (I): Both the robot turns till they are facing the goal. Then the robot starts moving towards the goal with a constant velocity. Similar to the Bug-1 both robots are idle till the ninth second because it is initially waiting for start command from the user. Then the velocity remains at 53 mm/s till it encounters an obstacle. This results in the sudden velocity drop near to the thirteenth second.

Section (II): The first robot acts as leader and starts circumnavigating the obstacle in clockwise direction. The other robot acts as follower and moves in the anticlockwise direction. The robots first orient themselves parallel to the obstacle and then start moving along the periphery of the obstacle till the corner is reached where it again lowers its velocity to about 35 mm/s while going around the corner as displayed by the velocity profile. After the corner, the robot continues moving with velocity of 53 mm/s. At regular time intervals both the robots calculate the distance to the goal at each point and the total distance travelled from the point of obstacle detection. The point having minimum distance to goal, distance to target at this point and total distance travelled by the robot is stored by each robot. Once the robot reaches within 30 mm from each other at twenty fifth second, the circumnavigation stops. The information stored by the robot is exchanged. The robot checks for the fastest route to

the point having minimum distance to the goal. The reason for the sudden dips in the velocity profile along the Section (II) is due to course correction as explained in the previous section.

Section (III): The robots meet each other. Since the shortest point is encountered by the follower robot, it turns 180° and move towards the shortest point. The leader robot waits for the follower to complete the turning and then continues in its path till point having minimum distance to goal is reached. The dip in the velocity profile at 29th second is due to this turning. The point having minimum distance to goal is reached at 34th second.

Section (IV): Once the robots reaches the point having minimum distance to goal, they orient themselves in the direction of the goal and starts to move in that direction while constantly checking the distance to the goal. The goal is considered attained if it is within 30 mm of either of the robots. During the experiments robot reached the goal, at thirty ninth second. The experimental results are consolidated in Table 2.

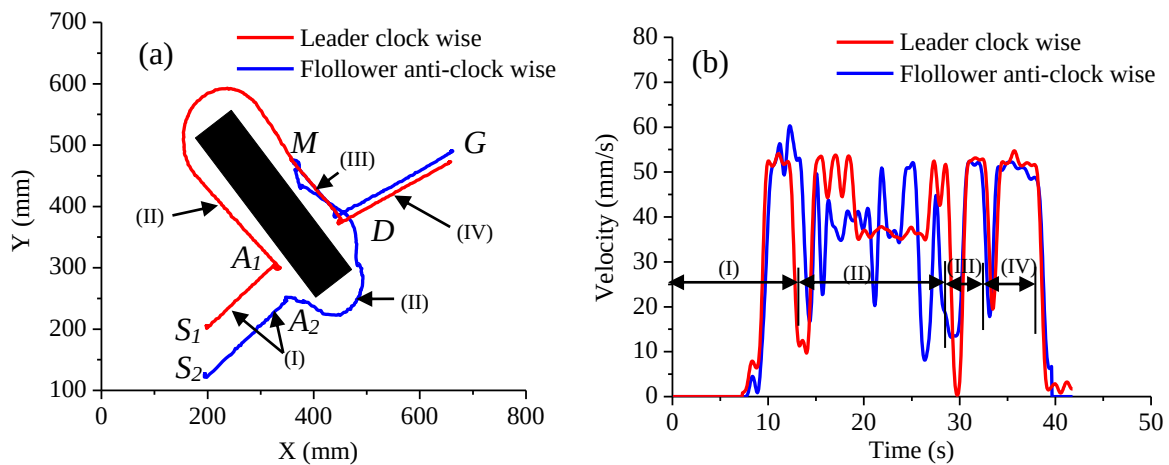


Fig. 13. New strategy (a) Trajectory of the robot (b) Velocity of the robot

Table 3. Experimental result

Number of robots	Time taken to reach the goal (s)	Average distance travelled by the robot (mm)	Net distance travelled by the robots (mm)
1	54	2440.7	2440.7
2	32	1195.05	2390.1

The total distance travelled by the robot while executing the new strategy is 2390 mm. The average distance travelled by the robot is 1195 mm which is 51 % less when compared to the distance travelled by the single robot executing Bug-1. This is because in the later the single robot has to circumnavigate the obstacle by itself while in the multi-robot system this distance is divided between two robots to cover. Also the point having minimum distance to goal is closer to the meeting point M in case of two robot system. This has reduced the distance to be travelled by the robot to reach the shortest point as against what happened in simulation. The total time taken by the robot to reach the goal is 32 s which is 44 % less than that required for the single robot system. When net distance travelled by the robot is considered, the multi robot system has 2% less net distance than the robot using Bug-1. The net distance would have been less for the Bug-1 system, if the robot was not taken longer path at top left corner as shown in Fig. 13 (a). The reason for this is explained in last part of this section.

Velocity plot of robot using Bug-1 strategy is layered with the velocity plot of robot using the modified strategy. Figure 14 describes this through a layered velocity plot. From the figure, it can be observed that the similar trends are there in both plots initially (plot of leader robot and plot of robot using the bug 1). It is to be observed that this is true till 35 s, where both the robot follow different path. Beyond 35 s, the robots using the modified algorithm move towards goal while the robot using the bug-1 continue circumnavigation. Hence, from 35 s onwards, the velocity plots are not comparable.

It should be noted here that, experiment 1 and experiment 2 are carried out two times. However, there was noticeable difference in the path tracked by the robot in each of these experiments. Each time, the robot is observed to take different path while circumnavigating near the corners of the obstacle. The major reason was adjudged to be the unpredictability in the output of infrared proximity sensor used in the robot. At corners of the obstacle, the

infrared sensor is found change in a random manner. This was not taken care during experiment. Implementation of better signal processing technique may solve this problem. So, the experimental results presented are a sample result to show the working of algorithm in a real scenario and for a general comparison.

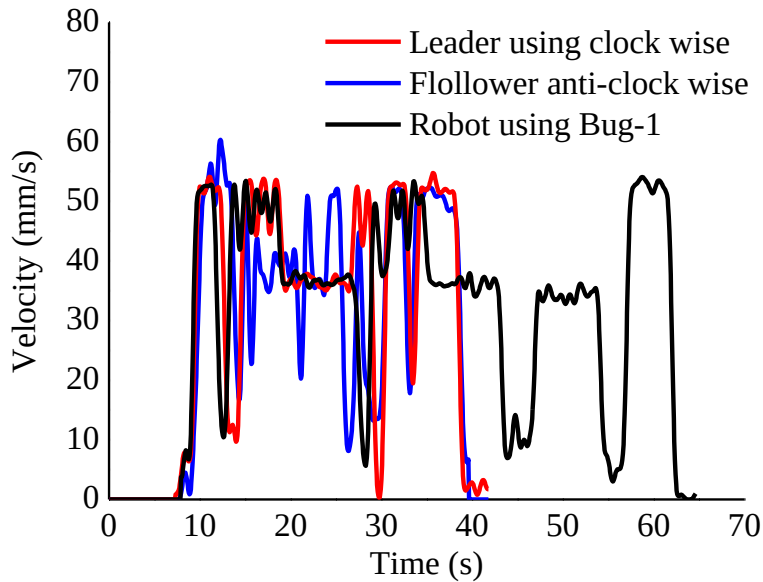


Fig. 14. Layered velocity plot

5.3 Comparison of experimental and simulation results

Simulation results in Table 1 shows that the average distance travelled by the robot is 1560 mm for Bug-1 and 1295 mm for new strategy. Whereas, experimental results show that the average distance travelled by the robot is 2440.7 mm in case of Bug-1 and 1195.05 in case of the new strategy. As mentioned in earlier section, the large difference in the experimental result of Bug-1 when compared to simulation result is due to robot taking a longer curve at both the corner due to incorrect signal from infrared proximity sensors. Similarly, the simulation results show travel time of 33 s for bug-1, and 25 s for new strategy. However, it is larger during experimentation in both for bug-1 and new strategy. There are two reasons for that. Firstly, the distance travelled by the robot during experimentation of Bug-1 was longer and secondly, the robot was programmed to reduce its velocity while navigating

around corners to reduce the effect of problem with the infrared sensors. Since the error between the experimental results and simulation results are majorly due to the factors which are not in the scope of current study, a fair comparison among them cannot be made in the current scenario.

6. Conclusions

A new multi-robot obstacle avoidance strategy has been developed based on Bug-1 algorithm. Theoretical study shows that the new strategy is better with respect to travel time and energy expenditure of robot. The result of simulation study as well as experimental study supports the theoretical study. Major conclusions of this work are listed below:

- From the theoretical analysis it is concluded that the travel time and the average distance covered by the robots in a multi-robot system executing the new strategy is always lower than that of a single robot executing the Bug-1 algorithm. The average energy expenditure of the robots executing the new strategy is also found to be lesser than that of the robot executing the Bug-1 algorithm
- Simulation results show that the new strategy outperforms Bug-1 algorithm irrespective of the number of robots involved in the system. The simulation results suggests that the improved nature of this algorithm makes it effective in multi-robot system
- The experiments carried out on e-puck robots supports the simulation results as well as theoretical analysis..
- A conclusive statement cannot be made about net energy usage as this can be larger or smaller for the new strategy depending on position of goal point, shape of obstacle, size of obstacle and number of robots etc. However, if the number of robots is equal in the multi robot system executing Bug-1 strategy and the new strategy, the net energy used and net distance travelled will be lower for the system executing the new strategy.

- A further argument for this strategy is that it performs better in many mazes where Bug-1 algorithms fail.

Following are some of the possibilities and potential areas for further improvement of the algorithm:

- Extend the algorithm to system consisting of large number of robots which exhibit swarm behavior. Extension of the new multi robot strategy for other obstacle avoidance methods derived from Bug-1 is an area to be explored.
- The motion of the robots in the current work is based on local sensing. However, a controller would be necessary for the robot to maintain formation.

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List of Notations

A, A_1, A_2	Initial points of contact of the robot with the obstacle
C	Midpoint of line connecting S_1 and S_2
D	Shortest distance point
d_{wh}	Diameter of the wheel
E_{avg-I}	Average of energy spend by the single robot systems
E_{avg-M}	Average of energy spend by the two robot system
$E_{total-I}$	Total energy expenditure for a single robot system
$E_{total-M}$	Total energy expenditure for a two-robot system
$E_{total-M1}$	Energy expenditure of leader robot in a two-robot system
$E_{total-M2}$	Energy expenditure of follower robots in a two-robot system
G	Goal point
H_1, H_2	Hitting points
L_1	Location having lowest distance to destination
L_2	Leave point
M	Meeting point
S, S_1, S_2	Starting positions of the robots
T_{1AA}	Time taken for the robot to go around the obstacle
T_{1AD}	The time required for the single robot to reach D from A
T_{2AM}	Time required for the two robots to meet at M after leaving A_1 and A_2
T_{2MD}	Time required for the robots to reach D from M
T_{DG}	Time required for all the robots to move from D to G
T_{SA}	Time taken for the robots to travel to initial point of contact
$T_{total-I}$	Total time taken by the single robot system
$T_{total-M}$	Time taken by the multi-robot system